Dynamic Programming Volumes I and II

A Quick Introduction — Lecture 2 of 2

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Review

In Lecture 1 we saw that standard DP theory does not apply in many settings

- discount rates are state-dependent
- no discounting & negative discounting
- nonlinear discounting
- nonlinear certainty equivalents / recursive preferences
- ambiguity and desire for robustness
- distributional DP
- nonstandard timing
- various combinations of these features, etc.

No general theory exists 😱

This motivates the **abstract dynamic programming** (ADP) framework developed in DP1–2

- Handles all cases in a unified way
- Provides conditions for optimality and convergence of algorithms

Abstraction has side benefits:

- Machine compatible — as discussed in Lecture 1
- Separates measure / functional analysis from foundational DP concepts, proofs
- Facilitates development of new results (machine assisted!)

To abstract and unify, let's **recall what's common** to these applications

1. The key primitives are the policy operators T_σ

All remaining objects can be defined from policy operators and sup operations

- Lifetime values $\{v_\sigma\}$ are the fixed points of $\{T_\sigma : \sigma \in \Sigma\}$
- Value function is defined by $v_\triangle := \sup_\sigma v_\sigma$
- Optimal policies are those satisfying $v_\sigma = v_\triangle$
- Bellman operator is defined by $T_\triangle v = \sup_\sigma T_\sigma v$
- v-greedy policies are those satisfying $T_\sigma v = T_\triangle v$

2. Policy operators are always order preserving

- $v \leqslant w \implies T_\sigma v \leqslant T_\sigma w$

Abstract Dynamic Programs

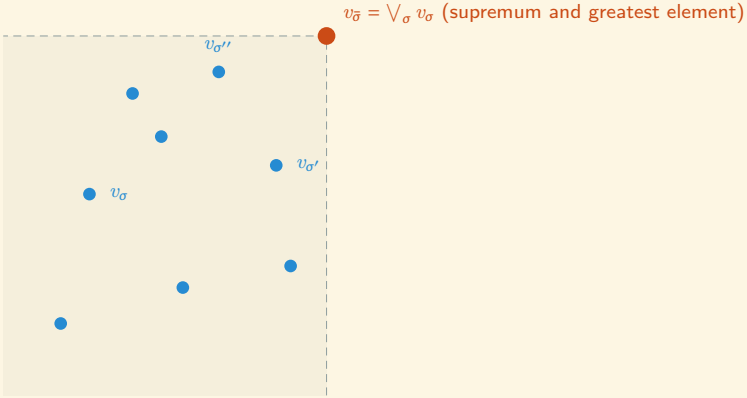
To define these objects we at least need

1. A set V for the v objects to live in
2. A set of maps T_σ that map V into V
3. A partial order $\preceq$ on V, to define order-preserving maps, suprema

In what follows,

$$\bigvee_\sigma v_\sigma := \text{ supremum of set } \{v_\sigma\} \text{ in } (V, \preceq)$$

Eg. Illustration of supremum $\bigvee_\sigma v_\sigma$ in $\mathbb{R}^2$



Definition 1

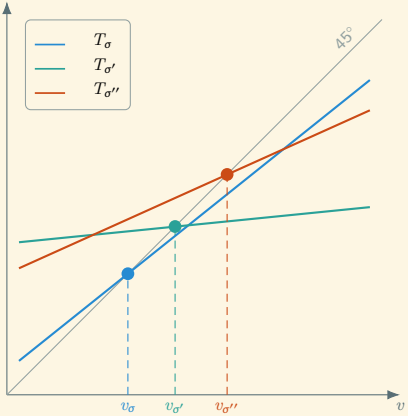
An **abstract dynamic program** (ADP) is a pair $(V, \mathbb{T})$, where

1. $V = (V, \preceq)$ is a partially ordered set and
2. $\mathbb{T} = \{T_\sigma : \sigma \in \Sigma\}$ is a family of order preserving self-maps on V

- V is called the **value space**
- Each T_σ is called a **policy operator**
- Each $\sigma \in \Sigma$ is called a **policy**

When each T_σ has a unique fixed point v_σ in V, we call the ADP **well-posed**

- v_σ is understood as representing "lifetime value" of σ



Eg. A well-posed ADP with three policies and $V \subset \mathbb{R}$

All the models we have seen so far can be represented as ADPs

Eg. Standard MDP

- Value space V is all bounded measurable $v\colon \mathsf{X} \to \mathbb{R}$ and $\preceq$ is pointwise partial order
- Policy operators take the form

$$(T_\sigma v)(x) = r(x, \sigma(x)) + \beta \int v(x') P(x, \sigma(x), \mathrm{d}x')$$

Eg. Distributional DP

- V = all conditional distributions $\eta(x, \cdot)$ with $x \in \mathsf{X}$, order $\preceq$ is pointwise FOSD
- Policy operators take the form $\eta \mapsto D_\sigma \eta$,

$$(D_\sigma \eta)(x, B) = \int \left[\int \mathbb{1}_B\{r_\sigma(x) + \beta v\} \, \eta(x', \mathrm{d}v) \right] P_\sigma(x, \mathrm{d}x')$$

ADP Theory

ADP definitions mimic definitions from applications:

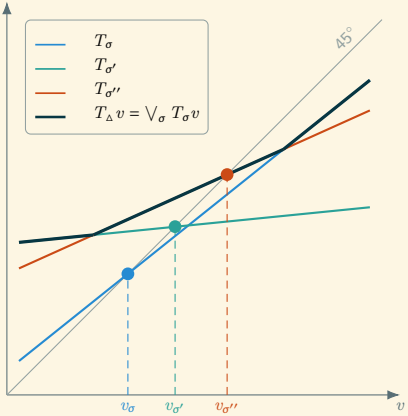
Letting $(V, \mathbb{T})$ be well-posed,

- the **Bellman operator** is defined by

$$T_\triangle v = \bigvee_{\sigma \in \Sigma} T_\sigma v$$

- We say that $v \in V$ satisfies the **Bellman equation** if

$$v = T_\triangle v$$



$T_\triangle$ is the upper envelope and $v_{\sigma''}$ satisfies the Bellman equation

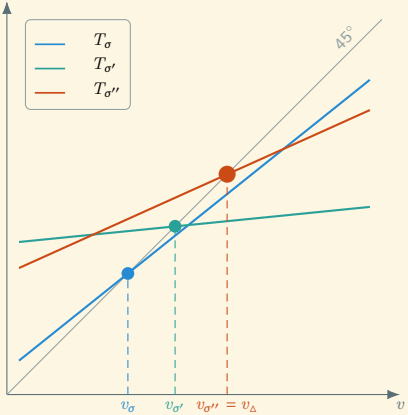
Also,

- the **value function** is defined by

$$v_\Delta = \bigvee_{\sigma \in \Sigma} v_\sigma$$

- a policy σ is called **optimal** if

$$v_\sigma = v_\Delta$$



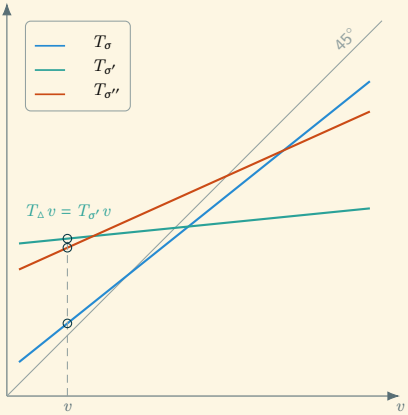
$v_\triangle = \bigvee_\sigma v_\sigma = v_{\sigma''}$, so σ'' is optimal

A policy $\sigma \in \Sigma$ is called **v-greedy** if

$$T_\sigma v = T_\Delta v$$

In addition, we define

- V_G := all $v \in V$ such that at least one v-greedy policy exists



σ' is v-greedy since $T_{\sigma'} v = T_\Delta v$

Let $(V, \mathbb{T})$ be a well-posed ADP

Definition 2

We say that the **fundamental optimality properties hold** for $(V, \mathbb{T})$ if

(B1) at least one optimal policy exists

(B2) v_Δ exists and is the unique solution to the Bellman equation in V_G

(B3) σ is optimal $\iff$ σ is v_Δ-greedy

When these properties hold we proceed as follows

1. Compute v_Δ by solving the Bellman equation
2. Compute a v_Δ-greedy policy — this is optimal

Lemma 1

For a well-posed ADP, (B2) implies both (B1) and (B3)

Sample Proof: [$v_\Delta \in V_G$ and v_Δ fixed under T_Δ] implies [every v_Δ-greedy σ is optimal]

- Since $v_\Delta \in V_G$ we can choose a v_Δ-greedy σ
- For this σ we have $T_\sigma v_\Delta = T_\Delta v_\Delta = v_\Delta$
- In particular, v_Δ is a fixed point of T_σ
- By well-posedness, the unique fixed point of T_σ is v_σ
- Hence $v_\sigma = v_\Delta$

Order Stability

When do the fundamental optimality properties hold?

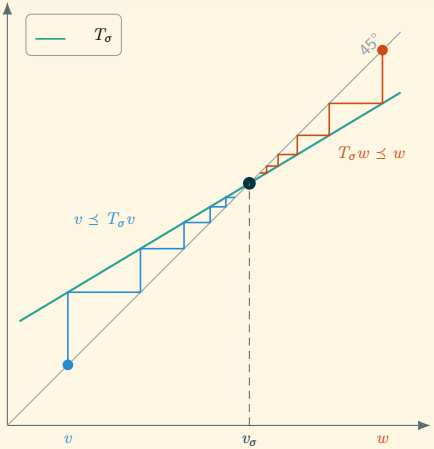
For this we need some kind of stability properties on the policy operators

In the traditional setting we have contraction properties

Order preserving + contractivity yields

$$v \leqslant T_\sigma v \text{ implies } T_\sigma^n v \uparrow v_\sigma \qquad \text{and} \qquad T_\sigma v \leqslant v \text{ implies } T_\sigma^n v \downarrow v_\sigma$$

- Iterates that start below the fixed point climb to it; those above descend



$v \leqslant T_\sigma v$ implies $T_\sigma^n v \uparrow v_\sigma$ and $T_\sigma v \leqslant v$ implies $T_\sigma^n v \downarrow v_\sigma$

Here we keep this as an **assumption**:

A well-posed ADP $(V, \mathbb{T})$ will be called **order stable** if each T_σ is order stable:

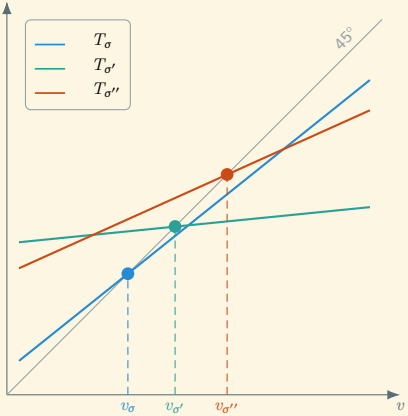
$$v \preceq T_\sigma v \text{ implies } T_\sigma^n v \uparrow v_\sigma \qquad \text{and} \qquad T_\sigma v \preceq v \text{ implies } T_\sigma^n v \downarrow v_\sigma$$

Now

- $\uparrow$ means sequence increasing and v_σ is the supremum
- $\downarrow$ means sequence decreasing and v_σ is the infimum

Same idea as before, but now on a poset $(V, \preceq)$

- Contraction not required — in fact no metric required



Eg. The simple three-policy ADP is order stable

Theorem 1: Max-Optimality

If $(V, \mathbb{T})$ is order stable, then the following statements are equivalent:

(i) T_Δ has a fixed point in V_G

(ii) The fundamental optimality properties hold

Proof: Omitted (see DP2)

Applications

Let's revisit some of the problems we looked at earlier

From Theorem 1, to show that the fundamental optimality properties hold, it suffices to show that

1. The policy operators are order stable
2. The Bellman operator has a fixed point that admits a greedy policy

Example: State-Dependent Discounting

Policy operators

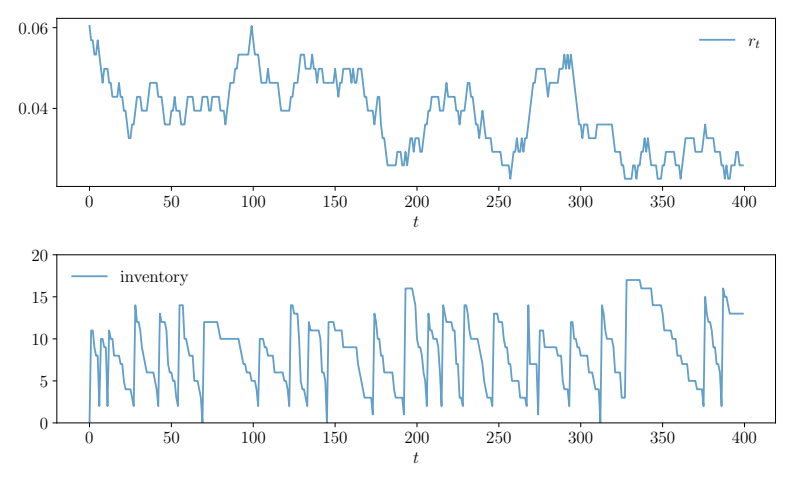
$$(T_\sigma v)(x, z) = \mathbb{E}\left[\pi(x, \sigma(x), D) + \beta(z) \int v(h(x, \sigma(x), D), z')\, Q(z, \mathrm{d}z')\right]$$

Under a spectral radius condition on $(Kf)(z) := \beta(z) \int f(z')\, Q(z, \mathrm{d}z')$,

- order stability holds on $V :=$ bounded measurable functions on $\mathsf{X} \times \mathsf{Z}$
- T_Δ has a fixed point in $V_G :=$ continuous functions in V

Hence fundamental optimality properties hold

VFI also converges on V_G


0.06
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Example: Inventory Problem with Risk Adjusted Expectations

$$(T_\sigma v)(x) = \mathscr{E}\left[\pi(x, \sigma(x), D) + \beta v(h(x, \sigma(x), D))\right]$$

and Bellman equation

$$v(x) = \max_{a \in \Gamma(x)} \mathscr{E}\left[\pi(x, a, D) + \beta v(h(x, a, D))\right]$$

Under mild conditions on $\mathscr{E}$, the ADP $(V, \mathbb{T})$ is order stable and T_Δ has a fixed point in V_G

Hence fundamental optimality properties hold

VFI also converges on V_G

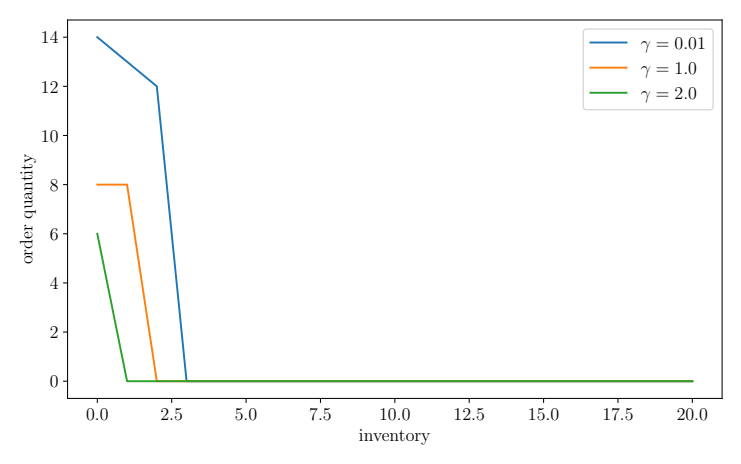


Figure 1: Risk-sensitive firms order less aggressively (Entropic certainty equivalent)

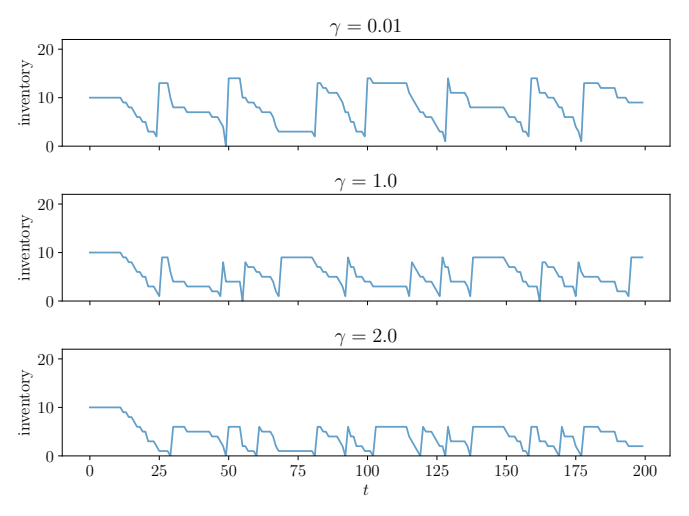


Figure 2: Inventory dynamics: more risk-sensitive firms hold less stock

Example: Irreversible Investment

Consider an irreversible investment problem with Bellman equation

$$v(k,z) = \max_i \left\{ f(k,z) - i + \beta \inf_{\psi \ll \varphi} \left[\int v(k', g(z,x))\psi(\mathrm{d}x) + \frac{1}{\gamma} D_{\mathrm{KL}}(\psi \,\|\, \varphi) \right] \right\}.$$

- $k' := i + (1-\delta)k$
- max is over i with $0 \leqslant i \leqslant \theta f(k,z)$

Interpretation: the manager does not fully trust the model

She fears misspecification in terms of the distribution φ of the shock sequence

We can show that the corresponding ADP is order stable and T_Δ has a fixed point in V_G

As a result, the fundamental optimality properties hold

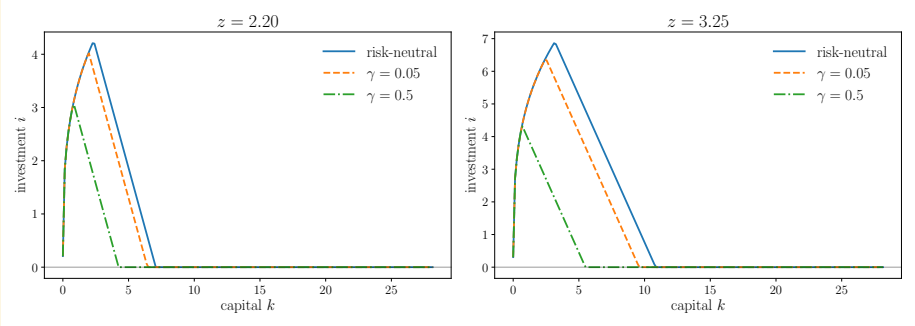


Figure 3: Investment policies under ambiguity aversion

Algorithms

Recall the big three algorithms: **VFI, OPI, and HPI**

These algorithms generalize to the abstract setting:

- **VFI:** iterate with T_Δ $\quad (v \mapsto T_\Delta v)$
- **OPI:** iterate with W $\quad (v \mapsto T_\sigma^m v$ where σ is v-greedy$)$
- **HPI:** iterate with H $\quad (v \mapsto v_\sigma$ where σ is v-greedy$)$

VFI: Iterating with $T_\triangle$

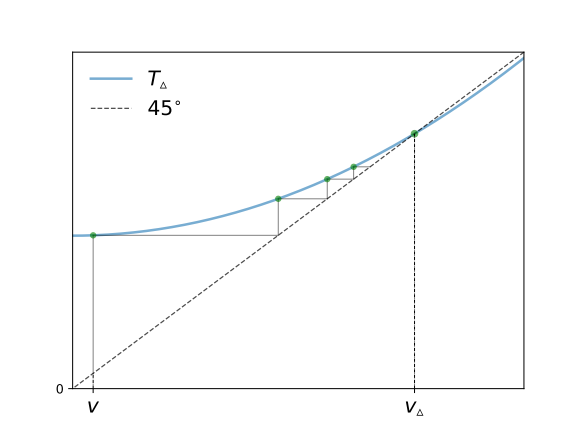


Figure 4: VFI: iterating with $T_\triangle$

HPI: Iterating with H

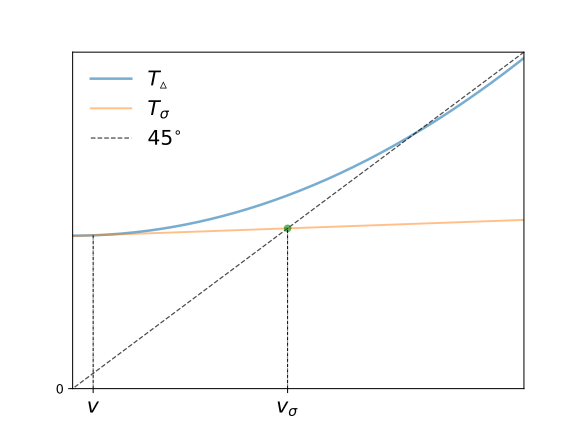


Figure 5: Choosing σ to be v-greedy implies $T_\sigma v = T_\Delta v$

HPI: Two Steps

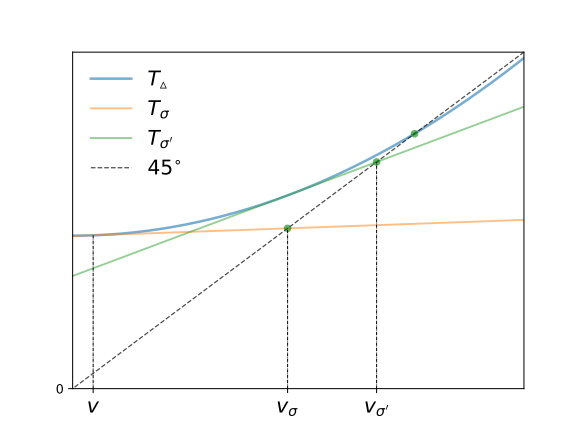


Figure 6: Two HPI steps: $v \to v_\sigma \to v_{\sigma'}$

Let $V_U = \{v \in V_G : v \preceq T_\Delta v\}$

We say that

- **VFI converges** if $T_\Delta^n v \uparrow v_\Delta$ for all $v \in V_U$
- **OPI converges** if $W^n v \uparrow v_\Delta$ for all $v \in V_U$
- **HPI converges** if $H^n v \uparrow v_\Delta$ for all $v \in V_U$

(Suprema, no topology)

Algorithms: Relationships

Proposition 1

Let $(V, \mathbb{T})$ be order stable with $V_G = V$. For all $n \in \mathbb{N}$ and v with $v \preceq T_\Delta v$,

$$v \preceq T_\Delta^n v \preceq W^n v \preceq H^n v \preceq v_\Delta$$

Remarks:

- HPI is "faster" than OPI, which is "faster" than VFI
- Proofs are simplified — VFI converges $\implies$ all converge

Aside: Which is actually faster?

It depends on the problem but also on changing software/hardware

VFI involves a large number of small steps (more sequential)

HPI involves a small number of large steps (more potential for parallelization)

OPI offers flexibility by varying m

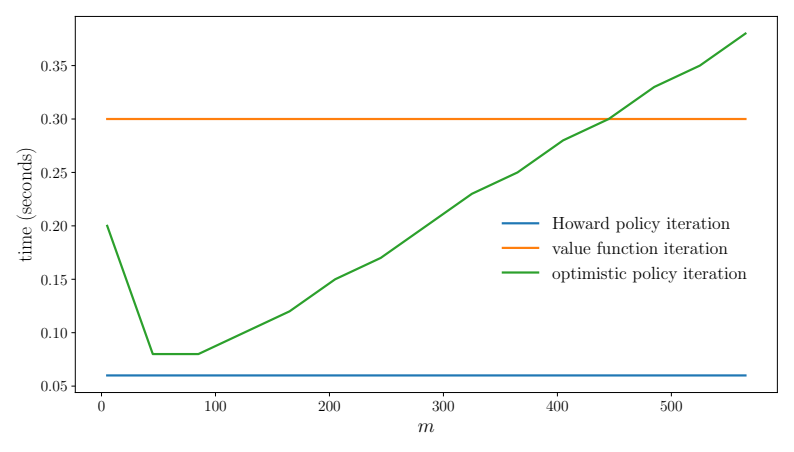


Figure 7: Solver times (JAX on GPU, finite states and actions)

Convergence Results for Algorithms

Now suppose that V has a topology as well as a (closed) partial order

We say that $(V, \mathbb{T})$ is **globally stable** if each $T_\sigma \in \mathbb{T}$ is globally stable:

$$T_\sigma \text{ has a unique fixed point } v_\sigma \in V \text{ and } T_\sigma^n v \to v_\sigma \text{ for all } v \in V$$

Theorem 2: Max-Optimality under Global Stability

Let $(V, \mathbb{T})$ be globally stable. If $V_G = V$ and T_Δ has a fixed point in V, then

1. the fundamental optimality properties hold and
2. VFI, OPI and HPI all converge

Example: A Real Option Problem

Launch product or wait, payoff $q(x)$ from launching in state x obeys

$$q(x) = \pi(x) + \beta(x) \int q(x')P(x, \mathrm{d}x')$$

Assume:

1. P has stationary distribution φ and $\pi \in V := L_1(\varphi)$
2. $\rho(K) < 1$, where K is the **discount operator** defined on V via

$$(Kv)(x) = \beta(x) \int v(x')P(x, \mathrm{d}x')$$

Can write post-launch profit as $q = (I - K)^{-1}\pi$

Policy operators:

$$(T_\sigma v)(x) = -c(x) + \int \beta(z)[\sigma(x')\, q(x') + (1 - \sigma(x'))v(x')]\, P(x, \mathrm{d}x')$$

Policies:

$$\Sigma := \text{ all measurable } \sigma\colon \mathsf{X} \to \{0, 1\}$$

Bellman operator:

$$(T_\vartriangle v)(x) = -c(x) + \beta(x) \int \max\{q(x'),\ v(x')\}\, P(x, \mathrm{d}x')$$

Can show: $(V, \mathbb{T})$ is globally stable and $T_\vartriangle$ has a fixed point in $V = V_G$

Hence fundamental optimality properties hold and VFI, HPI, and OPI all converge

Minimization

Definitions

- **Min-value function**: $v_\triangledown := \bigwedge_\sigma v_\sigma$
- **Min-Bellman operator**: $T_\triangledown\, v := \bigwedge_{\sigma\in\Sigma} T_\sigma\, v$
- σ is called **v-min-greedy** if $T_\sigma v = T_\triangledown\, v$
- σ is called **min-optimal** if $v_\sigma = v_\triangledown$

We obtain mirror images of our previous max-optimality results:

Theorem 3: Min-Optimality

If $(V, \mathbb{T})$ is order stable, then the following statements are equivalent:

(i) $T_\triangledown$ has a fixed point in $V_\triangledown^G$

(ii) The fundamental min-optimality properties hold

Theorem 4: Min-Optimality under Global Stability

Let $(V, \mathbb{T})$ be globally stable. If $V_G = V$ and $T_\triangledown$ has a fixed point in V, then

1. the fundamental min-optimality properties hold and
2. min-VFI, min-OPI and min-HPI all converge

Proofs

How to avoid re-doing proofs for the min case?

Key idea: infima in $(V, \preceq)$ correspond to suprema in the **order-dual** $V^\partial := (V, \preceq^\partial)$, where order is reversed

The **order-dual** of ADP $(V, \mathbb{T})$ is $(V^\partial, \mathbb{T})$

Steps

1. State hypotheses for min result in $(V, \mathbb{T})$
2. Infer corresponding properties in order dual $(V^\partial, \mathbb{T})$
3. Solve max problem in order-dual using previous max results
4. Shift results back to $(V, \mathbb{T})$

Example: Negative Discount Optimality

Recall the negative discounting problem with Bellman min-equation

$$v(x) = \min_{0 \leqslant a \leqslant x} \{c(a) + \delta\, v(x-a)\}$$

- $c(0) = 0,\ c' > 0,\ c'' > 0,\ \delta > 1$

Policy operators:

$$(T_\sigma v)(x) = c(\sigma(x)) + \delta\, v(x - \sigma(x))$$

Acting on V = all increasing $v\colon [0,1] \to \mathbb{R}$ with $v(0) = 0$

Since $\delta > 1$, these operators are not contractions

However, we can show that the fundamental min-optimality properties hold

Steps:

1. Pick a suitably restricted class of policies Σ
2. Show that T_σ is order stable on V for each σ in Σ
3. Show that $T_\triangledown$ has a fixed point in $v \in V$
4. Show that this fixed point has a v-min-greedy policy

Now apply Theorem 3

One consequence: We can construct an equilibrium for the production chain model from Lecture 1, where equilibrium prices obey

$$p(s) = \min_{0 \leqslant t \leqslant s} \{c(s-t) + \delta\, p(t)\}$$

(Value function of negative DP problem is an equilibrium price function)

FOC:

$$c'(s-t) = \delta\, p'(t)$$

Coase (1937): "a firm will tend to expand until the costs of organizing an extra transaction within the firm become equal to the costs of carrying out the same transaction by means of an exchange on the open market."

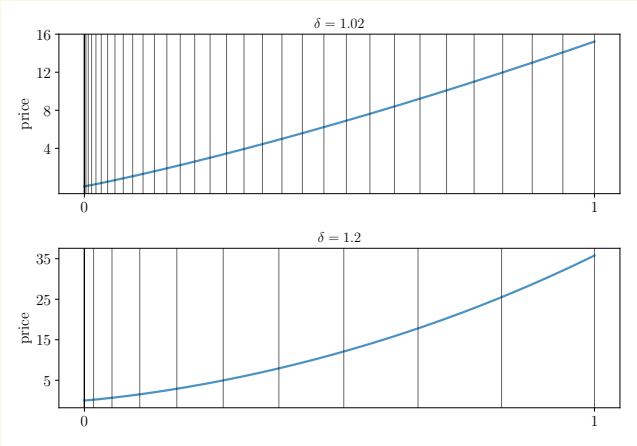


Figure 8: Equilibrium prices and firm boundaries for two values of δ

Example: Sequential Analysis

The problem can be reduce to an optimal stopping with policy operator

$$(T_\sigma g)(\pi) = \sigma(\pi)\, e(\pi) + (1 - \sigma(\pi)) \left[c + \int g(\pi')\, P(\pi, \mathrm{d}\pi') \right]$$

The Bellman min-operator is

$$(T_\triangledown g)(\pi) = \min \left\{ e(\pi),\ c + \int g(\pi')\, P(\pi, \mathrm{d}\pi') \right\}$$

Globally stable and $T_\triangledown$ has a fixed point in $V = V_\triangledown^G$

Hence fundamental min-optimality properties hold and min-algorithms converge

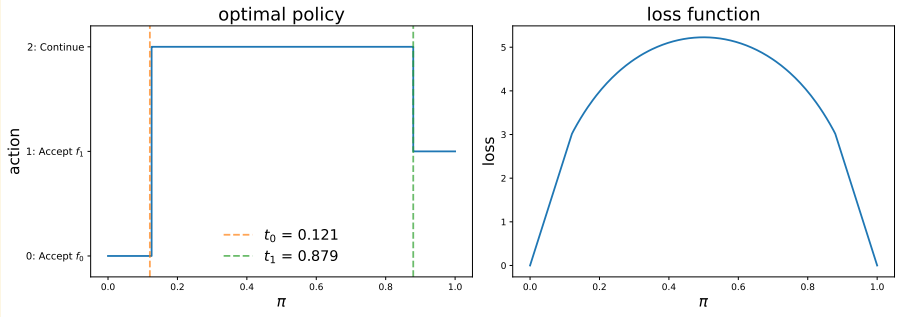


Figure 9: Optimal policy and loss function for the sequential analysis problem

Factorization

Some interesting ADPs come in pairs

Eg. MDP model with Bellman equation

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \int v(x') P(x, a, \mathrm{d}x') \right\}$$

The **Q-factor representation** (used for Q-learning) has Bellman equation

$$q(x, a) = r(x, a) + \beta \int \max_{a' \in \Gamma(x')} q(x', a') \, P(x, a, \mathrm{d}x')$$

Both can be represented as ADPs

What is the exact relationship between them?

Eg. Structural estimation

The **post-action value function** g satisfies

$$g(x, a) = \int \max_{a' \in \mathsf{A}} \left[r(x', a') + \beta g(x', a') \right] P(x, a, \mathrm{d}x')$$

This is obviously related to the traditional Bellman equation

$$v(x) = \max_{a \in \mathsf{A}} \left\{ r(x, a) + \beta \int v(x') P(x, a, \mathrm{d}x') \right\}$$

Both can be represented as ADPs

What is the exact relationship between them?

Eg. Recall the risk-sensitive inventory problem with Bellman equation

$$v(x) = \max_{a \in \Gamma(x)} \psi^{-1} \left\{ \mathbb{E}\, \psi \left[\pi(x, a, D) + \beta v(h(x, a, D)) \right] \right\},$$

where $\psi(t) = \exp(-\gamma t)$ with $\gamma > 0$

Let's say we want to do Q-learning with this model

Here's a guess of the Q-factor (shifts $\mathbb{E}$ to the outside)

$$q(x, a) = \mathbb{E}\left[\exp(-\gamma\, \pi(x, a, D)) \cdot \left(\min_{a' \in \Gamma(x')} q(x', a') \right)^{\beta} \right]$$

where $x' = h(x, a, D)$

Is this correct? Do these problems have the same optimal policies?

More generally

- How can we transform dynamic programs systematically?
- When are the fundamental optimality properties preserved?
- How are the optimal policies related?
- When is convergence of algorithms preserved?

Factored Dynamic Programs

The key idea: factor each policy operator into two maps composed in sequence

Eg. For a finite MDP with

$$(T_\sigma v)(x) = r(x, \sigma(x)) + \beta \int v(x')P(x, \sigma(x), \mathrm{d}x')$$

we can write $T_\sigma = G_\sigma \circ F$ where

- $F\colon v(x) \mapsto r(x, a) + \beta \int v(x')P(x, a, \mathrm{d}x')$
- $G_\sigma\colon g(x, a) \mapsto g(x, \sigma(x))$

Now consider iterating with $\hat{T}_\sigma = F \circ G_\sigma$ instead...

Since

- $F\colon v(x) \mapsto r(x,a) + \beta \int v(x')P(x,a,\mathrm{d}x')$
- $G_\sigma\colon g(x,a) \mapsto g(x,\sigma(x))$

setting $\hat{T}_\sigma = F \circ G_\sigma$ yields

$$(\hat{T}_\sigma q)(x,a) = r(x,a) + \beta \int q(x',\sigma(x'))\, P(x,a,\mathrm{d}x')$$

The Bellman equation for this new ADP is

$$q(x,a) = r(x,a) + \beta \int \max_{a' \in \Gamma(x')} q(x',a')\, P(x,a,\mathrm{d}x')$$

This is the Q-factor Bellman equation, for Q-learning!

In general, there are multiple ways that we can factor T_σ

- We'll see more examples below

And for complex ADPs, there can be a very large number of possible factorizations

This means that we have many ways to "rearrange" ADPs in possibly advantageous ways

What we want is

- Systematic ways to explore this
- Systematic ways to understand the relationships between such ADPs

FDPs

A **factored dynamic program** (FDP) is a tuple $(V, F, \hat{V}, \mathbb{G})$

- V and $\hat{V}$ are nonempty posets
- $F\colon V \to \hat{V}$ and $G_\sigma\colon \hat{V} \to V$
- $\{G_\sigma \hat{v}\}_{\sigma \in \Sigma}$ has a greatest element for every $\hat{v} \in \hat{V}$

Each FDP generates two ADPs

- **Primary ADP** $(V, \mathbb{T})$: $\quad T_\sigma = G_\sigma \circ F$
- **Subordinate ADP** $(\hat{V}, \hat{\mathbb{T}})$: $\quad \hat{T}_\sigma = F \circ G_\sigma$

We show: In many cases, this pair of ADPs share optimality properties

$$T_\sigma = G_\sigma \circ F \quad \text{(primary)}$$

$$\begin{tikzcd} V \arrow[r, "F"] \arrow[rr, bend left] & \hat{V} \arrow[r, "G_\sigma"] & V \end{tikzcd}$$

$$\hat{T}_\sigma = F \circ G_\sigma \quad \text{(subordinate)}$$

$$\begin{tikzcd} \hat{V} \arrow[r, "G_\sigma"] \arrow[rr, bend left] & V \arrow[r, "F"] & \hat{V} \end{tikzcd}$$

Example: Structural Estimation

The policy operators for the original MDP model take the form

$$(T_\sigma v)(x) = r(x, \sigma(x)) + \beta \int v(x')P(x, \sigma(x), \mathrm{d}x').$$

The post-action value function policy operators take the form

$$(\hat{T}_\sigma g)(x, a) = \int \left[r(x', \sigma(x')) + \beta g(x', \sigma(x'))\right] P(x, a, \mathrm{d}x')$$

These are primary and subordinate for the FDP with

- $F\colon v(x) \mapsto \int v(x')P(x, a, \mathrm{d}x')$ (state $\to$ state-action)
- $G_\sigma\colon g(x, a) \mapsto r(x, \sigma(x)) + \beta\, g(x, \sigma(x))$ (state-action $\to$ state)

Optimality for FDPs

Optimality can be studied via the (often simpler) subordinate ADP

Theorem 5: FDP Optimality (Order Preserving Case)

Let F and G_σ be order preserving

In this setting,

1. The fundamental optimality properties hold for the primary ADP if and only if they hold for the subordinate ADP
2. Any optimal policy for the primary ADP is optimal for the subordinate ADP
3. If $\hat{v}_\Delta$ is the value function of the subordinate ADP and $G_\sigma \hat{v}_\Delta = G_\Delta \hat{v}_\Delta$, then σ is optimal for the primary ADP

(Here G_Δ takes a sup over all G_σ)

Eg. To solve for an optimal policy for the original ADP

$$v(x) = \max_a \left\{ r(x,a) + \beta \int v(x') P(x,a,\mathrm{d}x') \right\}$$

we can solve the structural estimation Bellman equation

$$g(x,a) = \int \max_{a'} \left[r(x',a') + \beta g(x',a') \right] P(x,a,\mathrm{d}x')$$

to produce the fixed point g_Δ and then compute σ via

$$\sigma(x) \in \arg\max_a \{ r(x,a) + \beta\, g_\Delta(x,a) \}$$

This is a relatively easy case — we'll see a more complex one in the next section

Reinforcement Learning

Consider a finite MDP with Bellman equation

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \sum_{x'} v(x') P(x, a, x') \right\}$$

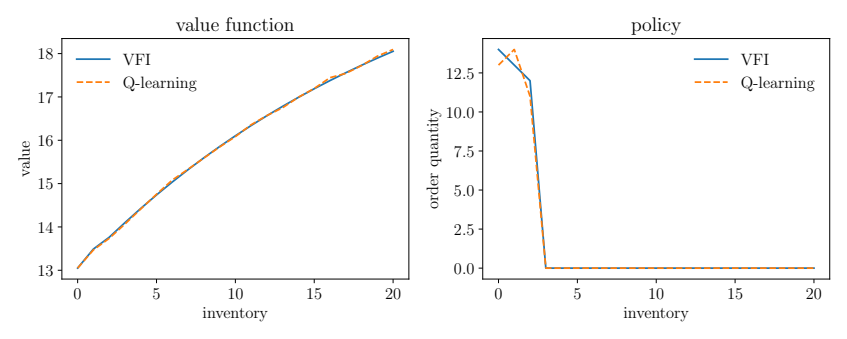
Reinforcement learning (RL) solves for optimal policies without knowledge of P, r

Learn the optimal Q-factor: update via

$$q_{t+1}(X_t, A_t) = (1 - \alpha_t)\, q_t(X_t, A_t) + \alpha_t \left[R_t + \beta \max_{a'} q_t(X_{t+1}, a') \right]$$

- Converges to the Q-factor fixed point q_Δ under standard conditions
- Optimal policy: $\sigma(x) = \arg\max_a q_\Delta(x, a)$

Example: Inventory problem (risk neutral)



Risk-Sensitive Q-Factor

Recall our discussion of the risk-sensitive inventory problem, which has policy operators

$$(T_\sigma v)(x) = \psi^{-1}\left\{\mathbb{E}\,\psi\left[\pi(x,\sigma(x),D) + \beta\, v(h(x,\sigma(x),D))\right]\right\}$$

If we set

$$(Fv)(x,a) := \mathbb{E}\exp[-\gamma\,(\pi(x,a,D) + \beta\; v(h(x,a,D)))]$$

and

$$(G_\sigma\, q)(x) := -\frac{1}{\gamma}\ln\big(q(x,\sigma(x))\big)$$

then the subordinate ADP has policy operators

$$(\hat{T}_\sigma q)(x,a) = \mathbb{E}\left[\exp(-\gamma\,\pi(x,a,D))\;\cdot\;q(h(x,a,D),\sigma(h(x,a,D)))^\beta\right]$$

Since $\gamma > 0$, both F and G_σ are order reversing:

$$(Fv)(x, a) := \mathbb{E} \exp[-\gamma\,(\pi(x, a, D) + \beta\, v(h(x, a, D)))]$$

and

$$(G_\sigma\, q)(x) := -\frac{1}{\gamma} \ln\big(q(x, \sigma(x))\big)$$

So the FDP max-optimality theorem (order-preserving case) does not apply!

We need an order-reversing version ...

FDP Min-Optimality

Theorem 6: FDP Optimality (Order Reversing Case)

Let F and G_σ be order reversing

In this setting,

1. The max-optimality properties hold for the primary ADP if and only if the min-versions hold for the subordinate ADP
2. Any max-optimal policy for the primary ADP is min-optimal for the subordinate ADP
3. If $\hat{v}_\nabla$ is the min-value function of the subordinate ADP and $G_\sigma \hat{v}_\nabla = G_\Delta \hat{v}_\nabla$, then σ is max-optimal for the primary ADP

This means we can compute the min-value function for the Q-factor version and use it to recover the optimal policy for the original problem

Risk-Sensitive Q-Learning

The Q-factor Bellman equation corresponding to the subordinate ADP is

$$q(x, a) = \mathbb{E}\left[\exp(-\gamma\, \pi(x, a, D)) \;\cdot\; \left(\min_{a' \in \Gamma(x')} q(x', a')\right)^{\beta}\right]$$

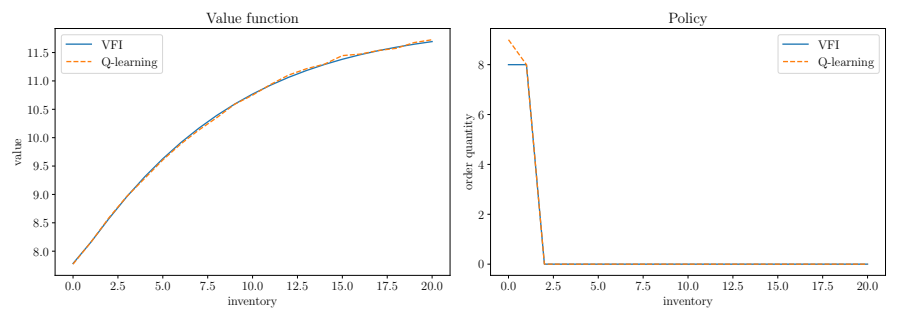
where $x' = h(x, a, D)$ — confirms our earlier guess

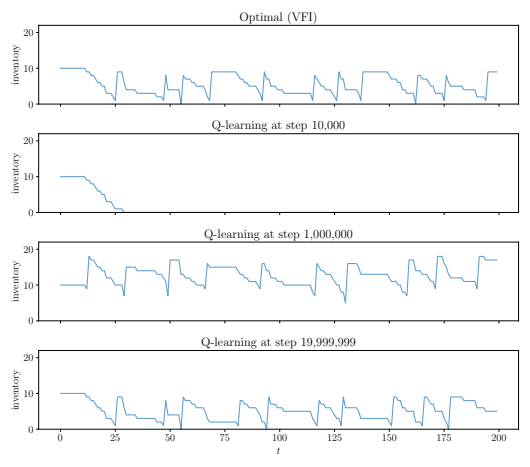
We can learn this risk-sensitive Q-factor from data

The Q-learning update rule becomes

$$q_{t+1}(x, a) = (1 - \alpha_t)\, q_t(x, a) + \alpha_t \left[\exp(-\gamma\, R_{t+1}) \;\cdot\; \left(\min_{a'} q_t(X_{t+1}, a')\right)^{\beta}\right]$$

Risk-sensitive Q-learning converges




Optimal (VFI)
Q-learning at step 10,000
Q-learning at step 1,000,000
Q-learning at step 19,999,999
inventory
0
10
20
0
25
50
75
100
125
150
175
200
t

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